

are also looking at launching manned missions into space eventually. But the closest to completion is not any nation – it comes from several privately owned companies.

SpaceShipOne

The Ansari X Prize was a competition organised by the X Prize Foundation and funded by Iranian entrepreneurs Amir Ansari and Anousheh Ansari – who was, incidentally, the world’s first self-funded space tourist. The competition was a challenge to a private organisation to come up with a viable reusable manned spacecraft that could be flown on two missions within a span of 14 days.

Burt Rutan, an American aerospace engineer who gained fame for the Voyager (the first aircraft to circumnavigate the globe without having to stop for fuel) took up the challenge with his firm Scaled Composites and won it with the SpaceShipOne in October 2004 when it crossed the 100-kilometre Karman line.

Unlike usual space launches, the SpaceShipOne wasn’t boosted into the air by a rocket. Instead, it was attached to a larger aircraft called the WhiteKnightOne, which took it into the skies and let it loose, whereupon it gathered wind and soared upwards.

SpaceShipOne is a three-place, high-altitude research rocket, designed for sub-orbital flights to 100 km altitude. The unique configuration allows aircraft-like qualities for boost, glide, and landing. The ship converts (via pneumatic-actuated feather) to a stable, high-drag shape for atmospheric entry. This “care-free” configuration allows a “hands-off” re-entry and greatly reduces aero/thermal loads.



The WhiteKnight aircraft carrying the SpaceShipOne before its further ascent into space